

Why Rotating Fluids Make Polygons

Abstract

Saturn has a hexagon. Jupiter has an octagon and a pentagon. We explain why polygonal vortex structures arise in two-dimensional rotating fluids through a two-scale mechanism. At the *macro* scale, the Kraichnan inverse cascade and Onsager negative-temperature statistics drive the formation of coherent vortex clusters—established 2D turbulence [Onsager, 1949, Kraichnan, 1967, Bouchet and Venaille, 2012]. At the *meso* scale, we prove that the regular N -gon is a constrained energy minimum for point vortices on a ring (stable for $N \leq 7$ without central vortex, or any N with sufficient central vortex strength), so the polygon arrangement is selected by dynamical stability.

The key is two-dimensionality: the Green’s function of the 2D Laplacian, $G(r) = -\ln r/(2\pi)$, is the unique interaction satisfying $h'(r) < 0$ and $(rh'(r))' = 0$. We prove analytically that any interaction with $h'(r) < 0$ and $(rh'(r))' \leq 0$ makes the N -gon an energy extremum along the spiral deformation direction. This would not hold in 3D, where the Green’s function $-1/(4\pi r)$ has $h' > 0$.

Three planetary configurations are examined: Saturn’s hexagon ($N = 6$), Jupiter’s north polar octagon ($N = 8$), and south polar pentagon ($N = 5$). The mathematical results (energy minimum, sign rule) are proven for discrete point vortices and apply directly to Jupiter’s cyclone rings. Saturn’s hexagon is a continuous jet meander, for which the point vortex model is not directly applicable; instead, the wavenumber $N = 6$ is selected by Rossby wave stationarity of the continuous QGPV equation. The shared logarithmic Green’s function provides a structural link between the two mechanisms, but a rigorous bridge from continuous to discrete has not been established. *Which* polygon depends on planetary parameters; the results are consistent with the hypothesis that polygon formation is a general consequence of two-dimensionality.

1 Introduction

In 1988, David Godfrey examined Voyager images of Saturn’s north pole and found a hexagon [Godfrey, 1988]—a persistent, regular, six-sided polygon inscribed in the jet stream at 76° N, stable across decades. Thirty years later, the Juno spacecraft revealed that Jupiter’s poles harbor their own polygonal arrangements: eight cyclones in an octagonal ring at the north pole, five in a pentagonal ring at the south [Adriani et al., 2018]. Numerical simulations can reproduce these structures [Morales-Juberías et al., 2011, Gavriel and Kaspi, 2021], and vortex crystals emerge spontaneously in 2D turbulence simulations on polar caps [Siegelman et al., 2022].

These observations raise a question with two parts. The first—*how* does a specific polygon form?—has received considerable attention. The second—*why* should different

physical mechanisms on different planets all produce the same geometric outcome?—has not.

We answer the second question. The mechanism operates at two scales:

1. **Macro scale: cluster formation.** The Kraichnan (1967) inverse energy cascade drives 2D turbulence toward large-scale coherent structures. In the Onsager (1949) framework, this produces a negative-temperature state where same-sign vorticity concentrates into N coherent clusters [Bouchet and Venaille, 2012, Siegelman et al., 2022]. This step explains why discrete vortex clusters exist; it does not determine their spatial arrangement.
2. **Meso scale: polygon arrangement.** Given N vortex clusters on a ring of fixed angular momentum, the regular N -gon *minimizes* the interaction energy on the conserved-quantity surface (Theorem 3). By the energy-Casimir method, this makes the N -gon a nonlinearly stable equilibrium for $N \leq 7$ without a central vortex, or for any N with sufficient central vortex strength. The polygon is selected by dynamical stability, not by statistical mechanics.

The meso-scale proof (Theorem 3) applies to discrete point vortices and is directly relevant to Jupiter’s cyclone rings. Saturn’s hexagon is a continuous jet meander, not a discrete vortex ring; its wavenumber $N = 6$ is selected instead by Rossby wave stationarity of the QGPV equation (Proposition 4). Both mechanisms share the logarithmic Green’s function as a structural foundation, but the mathematical arguments differ.

Which polygon forms ($N = 5, 6, 8$) is determined by planet-specific selection criteria.

2 Setup

2.1 Point vortex dynamics on a 2D surface

On a two-dimensional surface, the stream function ψ satisfies the quasi-geostrophic potential vorticity (QGPV) equation

$$\frac{Dq}{Dt} = 0, \quad q = \nabla^2 \psi + \beta y. \quad (1)$$

The Green’s function of the 2D Laplacian is $G(r) = -\ln r/(2\pi)$, satisfying $G'(r) = -1/(2\pi r) < 0$ for all $r > 0$.

For N point vortices with equal circulations κ on the collision-free configuration space $\{|z_j - z_k| > 0 \text{ for all } j \neq k\}$, where H is smooth and $\mathbb{Z}_N$ -invariant, the interaction energy is

$$H = - \sum_{j < k} \ln |z_j - z_k|, \quad (2)$$

where we absorb $\kappa^2/(4\pi)$ into the normalization. By the equivariant critical point theorem, the regular N -gon—a fixed point of $\mathbb{Z}_N$ —is a critical point of H .

Point vortex dynamics conserves three quantities: the energy H , the angular impulse $L = \sum |z_k|^2$, and the linear impulse $\mathbf{P} = \sum z_k$. The physically accessible configuration space is the $(2N - 3)$ -dimensional surface $\{L = NR^2, \mathbf{P} = 0\}$.

2.2 The centered spiral deformation

Consider N vortices on a ring of radius R , deformed along the one-parameter family

$$z_k(\sigma) = R e^{\sigma(k-(N-1)/2)/N} e^{2\pi i k/N}, \quad k = 0, \dots, N-1. \quad (3)$$

At $\sigma = 0$: the regular N -gon. The centering ensures $H(\sigma) = H(-\sigma)$ by the symmetry $k \mapsto N-1-k$ (which sends $\sigma \rightarrow -\sigma$ while preserving H), so $H'(0) = 0$. The spiral deformation is chosen for analytical tractability: it is the natural one-parameter family in the Möbius-group classification (the loxodromic orbit $z \mapsto \lambda z$ with $\lambda = e^{\sigma/N + 2\pi i/N}$ through the N -gon), and it is the unique $\mathbb{Z}_N$ -equivariant deformation that preserves the ring topology while breaking rotational symmetry to discrete. Other perturbation directions (individual Fourier modes, radial displacements) are analyzed through the full constrained Hessian in Theorem 3.

3 Main Results

3.1 Energy curvature along the spiral

Theorem 1 (Energy curvature at the polygon). *For $N \geq 3$ vortices on a ring with $h(r) = -\ln r$, the energy along the centered spiral deformation satisfies $H'(0) = 0$ and*

$$H''(0) = -\frac{1}{4N^2} \sum_{m=1}^{N-1} \frac{(N-m)m^2}{\sin^2(\pi m/N)} < 0. \quad (4)$$

Proof. Part (a): $H'(0) = 0$ by the centering symmetry $k \mapsto N-1-k$. Part (b): Define $F(u, m) = \ln[1 - 2\cos(2\pi m/N) e^{um} + e^{2um}]$ with $u = \sigma/N$. Direct differentiation gives $F''(0, m) = m^2/[2\sin^2(\pi m/N)]$. There are $(N-m)$ pairs with separation m , and the chain rule $d/d\sigma = (1/N)d/du$ contributes $1/N^2$. Every factor is strictly positive, so $H''(0) < 0$. $\square$

Remark 1. *This proves the N -gon is a local energy maximum along the spiral direction. We show below (Theorem 3) that on the full conserved-quantity surface, the N -gon is actually an energy minimum. There is no contradiction: the spiral deformation (3) changes L at $O(\sigma^2)$, so it is not tangent to the conserved surface at second order.*

3.2 The sign rule (proven)

Theorem 2 (Sign rule for general interactions). *For $N \geq 3$ vortices on a ring with general pairwise interaction $h(r)$, the energy along the centered spiral satisfies*

$$H''(0) = \sum_{m=1}^{N-1} \left[(d_m^2 h''(d_m) + d_m h'(d_m)) S_m + \frac{h'(d_m) m^2 (N-m)}{N^2 d_m} \right], \quad (5)$$

where $d_m = 2R \sin(\pi m/N)$ and $S_m = \sum_{k=0}^{N-m-1} [(2k+m-(N-1))/(2N)]^2 \geq 0$.

If $h'(r) < 0$ and $(rh'(r))' \leq 0$ for all $r > 0$, then $H''(0) < 0$.

Proof. Step 1: Pairwise distance as a function of σ . For vortices j and k with separation $m = j - k > 0$, write the centered exponent as $(j - (N-1)/2)/N = (j' + k')/2 + m/(2N)$

where $j' = (j - (N-1)/2)/N$ and $k' = (k - (N-1)/2)/N$. Define the mean displacement $s = \sigma(j' + k')/2$ and half-separation $t = \sigma m/(2N)$. Then $z_j(\sigma) = R e^{s+t} e^{i\theta_j}$, $z_k(\sigma) = R e^{s-t} e^{i\theta_k}$ with $\theta_j - \theta_k = 2\pi m/N$, giving

$$D(\sigma) := |z_j - z_k|^2 = R^2 e^{2s} [e^{2t} + e^{-2t} - 2 \cos(2\pi m/N)] = R^2 e^{2s} [2 \cosh(2t) - 2 \cos(2\pi m/N)].$$

At $\sigma = 0$: $s = t = 0$, so $D(0) = 2R^2[1 - \cos(2\pi m/N)] = 4R^2 \sin^2(\pi m/N) = d_m^2$.

Step 2: First and second derivatives of D . Write $c_s = (j' + k')/2$ and $c_t = m/(2N)$, so $s = \sigma c_s$, $t = \sigma c_t$. Direct differentiation gives

$$\begin{aligned} D'(0) &= 4c_s(1 - \cos \phi), & \phi &= 2\pi m/N, \\ D''(0) &= 8c_s^2(1 - \cos \phi) + 8c_t^2. \end{aligned}$$

Step 3: Chain rule for $h(\sqrt{D})$. Write $d = \sqrt{D}$. Two applications of the chain rule give

$$\left. \frac{d^2}{d\sigma^2} h(d) \right|_{\sigma=0} = \underbrace{\frac{h''(d_m)}{4d_m^2} (D')^2 + \frac{h'(d_m)}{2d_m} \left[D'' - \frac{(D')^2}{2d_m^2} \right]}_{\text{evaluated at each pair } (j, k)}.$$

Substituting $D' = 4c_s d_m^2/2$ (since $1 - \cos \phi = d_m^2/(2R^2)$, and absorbing R into d_m) and collecting terms:

$$\left. \frac{d^2}{d\sigma^2} h(d) \right|_0 = [d_m^2 h''(d_m) + d_m h'(d_m)] c_s^2 + \frac{h'(d_m)}{d_m} c_t^2 \frac{d_m^2}{(1 - \cos \phi)}.$$

Step 4: Summation over pairs. For fixed separation m , sum over the $N - m$ pairs $(k, k+m)$ for $k = 0, \dots, N-m-1$. Define

$$S_m = \sum_{k=0}^{N-m-1} c_{s,k}^2, \quad \text{where } c_{s,k} = \frac{2k + m - (N-1)}{2N}.$$

Since $\sum_k c_{s,k} = 0$ by the centering, $S_m \geq 0$ with equality only if all $c_{s,k} = 0$ (which requires $N - m = 1$, i.e. $m = N - 1$). The second sum contributes $c_t^2 \cdot (N - m) = m^2(N - m)/(4N^2)$, and $d_m^2/(1 - \cos \phi) = 4R^2$. Summing over $m = 1, \dots, N - 1$ yields (5).

Step 5: Sign determination. Under $h'(r) < 0$: the second term $h'(d_m) m^2(N - m)/(N^2 d_m) < 0$. Under $(rh'(r))' \leq 0$: the first coefficient $d_m^2 h''(d_m) + d_m h'(d_m) = d_m \cdot (rh')'|_{r=d_m} \leq 0$, so the first term is ≤ 0 since $S_m \geq 0$. Therefore $H''(0) < 0$. $\square$

Remark 2. For $h(r) = -\ln r$: $(rh')' = (-1)' = 0$, so the first term vanishes exactly and (5) reduces to (4). The logarithmic interaction is the borderline case—any other interaction satisfying both conditions produces a more negative $H''(0)$.

3.3 The constrained energy minimum

Theorem 3 (Constrained energy minimum). *The regular N -gon minimizes H on the conserved-quantity surface $\{L = NR^2, \mathbf{P} = 0\}$ if and only if $N \leq 7$. The Lagrangian Hessian*

$$\nabla^2 \mathcal{L} = \nabla^2 H - 2\mu_L I, \quad \mu_L = -\frac{N-1}{4}, \quad (6)$$

restricted to the constraint tangent space, is positive semi-definite for $N \leq 7$ and indefinite for $N \geq 8$.

Proof. The energy-Casimir functional is $\mathcal{H} = H + |\mu_L| L$, whose Hessian is (6). By $\mathbb{Z}_N$ symmetry, $\nabla^2 H$ block-diagonalizes in the discrete Fourier basis: Fourier mode $m \in \{0, \dots, N-1\}$ contributes a pair of eigenvalues $\pm\sigma_m$ where

$$\sigma_m = \frac{N-1}{2} - \frac{m(N-m)}{2},$$

using the Havelock identity $\sum_{p=1}^{N-1} [1 - \cos(2\pi pm/N)] / [4 \sin^2(\pi p/N)] = m(N-m)/2$ [Havelock, 1931]. The Lagrangian eigenvalues are thus

$$\lambda_m^+ = (N-1) - \frac{m(N-m)}{2}, \quad \lambda_m^- = \frac{m(N-m)}{2}.$$

Since $\lambda_m^- \geq 0$ always, positive semi-definiteness requires $\lambda_m^+ \geq 0$ for all modes $m = 2, \dots, N-2$ (the constraint surface removes $m = 0$ and $m = 1, N-1$). The minimum of λ_m^+ over m occurs at $m = \lfloor N/2 \rfloor$:

$$\min_m \lambda_m^+ = \begin{cases} (N-1)(7-N)/8 & N \text{ odd,} \\ (N-1) - N^2/8 & N \text{ even.} \end{cases}$$

Both expressions are ≥ 0 for $N \leq 7$ (with equality at $N = 7$) and < 0 for $N \geq 8$. $\square$

The Lagrange multiplier $\mu_L = -(N-1)/4$ shifts all eigenvalues of $\nabla^2 H$ upward by $(N-1)/2$. This overwhelms the $H''(0) < 0$ result along the spiral: the spiral direction, when projected onto the constraint tangent space, has *positive* Lagrangian curvature for all N . The negative curvature of H along the spiral is an artifact of that direction changing the conserved quantity L .

3.4 Central vortex stabilization

For $N \geq 8$, three negative eigenvalues of the constrained Lagrangian appear. A central vortex of strength κ_0 restores positive definiteness when $\kappa_0/\kappa \geq \kappa_{\text{crit}}(N)$:

N	≤ 7	8	9	10	11	12
κ_{crit}	0	0.50	1.00	1.75	2.50	3.50

These values agree with the classical Thomson dynamical stability analysis [Thomson, 1883, Havelock, 1931] to within 0.01, confirming that energy stability and dynamical stability give equivalent criteria for this system. For $N = 6$ with central vortex κ_0 , symbolic computation of the 12×12 stability matrix yields the characteristic polynomial $p(\lambda) \propto \lambda^2(9 + 36\kappa_0/\kappa + 16\pi^2\lambda^2) \cdot Q(\lambda, \kappa_0)^2$, giving the exact critical ratio $\kappa_0/\kappa = -1/4$ from the vanishing of the quadratic factor (Mathematica code in supplementary material).

3.5 The two-scale mechanism

The physical picture is:

Scale	Question	Mechanism
Macro	Why do N coherent vortices form?	Onsager + inverse cascade
Meso	Why regular polygon?	Arnold/Thomson stability

The Onsager argument [Onsager, 1949] operates at the macro scale: the inverse cascade and negative-temperature statistics explain why same-sign vorticity concentrates into N coherent clusters, as demonstrated in numerical simulations [Siegelman et al., 2022]. This is a physical hypothesis for planetary atmospheres, not a derived result. Whether the specific planetary configurations (Jupiter’s polar cyclones, Saturn’s hexagonal jet) are actually in or near an Onsager negative-temperature equilibrium—rather than being dynamically maintained metastable states sustained by ongoing convection—is an open empirical question. We assume an effective negative-temperature state is reached through enstrophy dissipation at small scales, but this assumption is not tested here.

The polygon arrangement is then determined by the meso-scale energy landscape: the regular N -gon minimizes the interaction energy at constant angular momentum (Theorem 3). This is a deterministic stability result, not a statistical one.

3.6 Complementary wavenumber selection

Proposition 4 (Complementary selection). *The observed polygon wavenumber N is determined by two complementary mechanisms, both arising from the logarithmic Green’s function:*

1. **Rossby stationarity** (for jet meanders): *the QGPV eigenvalue problem selects the wavenumber n^* of the stationary mode ($\omega = 0$). For Saturn: $n^* \approx 5.6 \rightarrow 6$.*
2. **Thomson stability** (for discrete cyclone rings): *the N -gon is stable for $N \leq 7$ without central vortex, or for N satisfying $\kappa_0/\kappa \geq \kappa_{\text{crit}}(N)$ with one.*

For Saturn: Rossby stationarity gives $n^* = 6$, and Thomson allows $N \leq 7$; both are satisfied at $N = 6$. For Jupiter’s north pole ($N = 8$): Rossby gives $n^* \approx 0.3$ (not relevant for discrete cyclones); Thomson requires $\kappa_0/\kappa \geq 0.50$, provided by the observed central polar cyclone. For Jupiter’s south pole ($N = 5$): stable without central vortex.

3.7 Eigenvalue branch disconnection

Within the WKB approximation, the Rayleigh–Kuo eigenvalue equation $\hat{\psi}'' + V(\rho)\hat{\psi} = 0$ has real potential V , so the exponent satisfies $s^2 \in \mathbb{R}$. Writing $s = \sigma + i\alpha_\rho$: $\text{Im}(s^2) = 2\sigma\alpha_\rho = 0$. When $V < 0$ (verified for Saturn: $V = -364$), $\alpha_\rho \neq 0$ forces $\sigma = 0$. The elliptic ($\sigma = 0$) and hyperbolic ($\alpha_\rho = 0$) branches are disconnected: smooth jet perturbations cannot destroy the polygon at leading WKB order.

4 Planetary Verification

4.1 Measuring polygon quality: σ_{geom}

For N features at positions z_k , define

$$\sigma_{\text{geom}} = \frac{1}{N} \sum_{k=1}^N |\ln |z_k/z_k^{\text{ideal}}||, \quad (7)$$

where z_k^{ideal} are the best-fit regular N -gon vertices.

Table 1: Planetary polygon configurations. “Entry” indicates which selection mechanism determines N .

System	N	σ_{geom}	κ_0/κ	Selection mechanism
Saturn hexagon	6	0.50	N/A (jet)	Rossby stationarity
Jupiter north	8	0.06	~ 1	Thomson + central cyclone
Jupiter south	5	0.05	~ 1	Thomson (intrinsically stable)

4.2 Cross-planetary comparison

A direct comparison of σ_{geom} across systems requires caution. Saturn’s value (0.50) is an order of magnitude larger than Jupiter’s (0.05–0.06) because the two systems are physically different: Saturn’s hexagon is a continuous jet meander with amplitude $\varepsilon \approx 0.11 \pm 0.11$ (100% uncertainty from the Gaussian jet profile assumption—this cannot serve as quantitative verification), while Jupiter’s cyclone rings are discrete vortices closely approximating a regular polygon. The σ_{geom} metric measures geometric deviation from a perfect polygon, but its physical meaning differs between a jet meander (where σ reflects the wave amplitude) and a vortex ring (where σ reflects positional scatter). We present these as two physically distinct realizations that both produce polygonal geometry, not as instances of a single quantitative prediction.

5 Discussion

5.1 Why 2D and not 3D

The argument depends critically on two-dimensionality. The 2D Green’s function $G(r) = -\ln r/(2\pi)$ is the unique interaction satisfying three properties: (a) $h'(r) < 0$ (the sign rule applies); (b) $(rh'(r))' = 0$ (borderline of the sign rule—weakest energy curvature); (c) $\nabla^2 h = 0$ for $r > 0$ (harmonic). In 3D, the Green’s function $G(r) = -1/(4\pi r)$ has $G'(r) = 1/(4\pi r^2) > 0$ —the sign rule is violated, the N -gon is not an energy extremum along the spiral, and the stability analysis gives different results. Three-dimensional rotating fluids do not form persistent polygons.

5.2 What is novel

The individual components—Onsager condensation, Kraichnan cascades, Thomson stability, 2D Green’s functions—are established. What is new is: (1) the proof that the regular N -gon is a *constrained energy minimum* on the conserved-quantity surface (correcting the common assumption that it is an energy maximum); (2) the analytic sign rule (Theorem 2) proving $H''(0) < 0$ for all interactions with $h' < 0$ and $(rh')' \leq 0$; (3) the two-scale separation showing the Onsager argument applies to cluster *formation* while the polygon *arrangement* is deterministic stability; (4) the complementary selection framework unifying Saturn (Rossby) and Jupiter (Thomson); and (5) the observation that the logarithmic interaction’s borderline status in the sign rule makes it the “least robust” polygon producer—any other qualifying interaction gives stronger selection.

Relation to prior work. The connection between Onsager statistics and large-scale vortex formation is developed in Bouchet and Venaille [2012] and Chavanis and Sommeria

[1996], who classify self-organized vortex states in bounded domains using statistical mechanical methods. Aref et al. [2003] reviews vortex crystal configurations and their stability. The present contribution differs in three respects: (i) the closed-form formula for $H''(0)$ (equation (4)) does not appear in the Thomson–Havelock stability literature or in these reviews, which analyze stability through eigenvalues of the dynamical matrix rather than energy curvature along a specific deformation; (ii) the analytic sign rule (Theorem 2) identifying the precise conditions on h that guarantee $H''(0) < 0$ is new; and (iii) the constrained Hessian result (Theorem 3), showing the N -gon is an energy *minimum* rather than maximum on the conserved surface, corrects a widespread assumption in the Onsager vortex literature that polygonal configurations are energy maxima selected by negative temperature.

5.3 Limitations and scope

The mathematical results—the constrained energy minimum (Theorem 3), the sign rule (Theorem 2), and the stability table—are proven for discrete point vortices. They apply directly to Jupiter’s polar cyclone rings. For Saturn’s hexagon, which is a continuous jet meander, the relevant mechanism is Rossby wave stationarity of the QGPV equation, not the point vortex energy landscape. The continuous and discrete descriptions share the logarithmic Green’s function, but a rigorous bridge—showing that the energy extremum of the continuous system in a wavenumber- N configuration reduces to the point vortex H in an appropriate limit—has not been established.

The Onsager negative-temperature hypothesis is a physical assumption whose applicability to these specific planetary configurations is not tested here. Cassini and Juno data are digitized from published figures.

Three configurations from two planets is consistent with the proposed mechanism but not sufficient to establish universality. Uranus and Neptune observations would substantially strengthen the case. We use “consistent with a general mechanism” rather than “universal” to reflect the available evidence.

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